I. Genetic Algorithm

Pseudo-code of Genetic Algorithm (GA)

Inputs

$\boldsymbol{T}$: max iteration, $\boldsymbol{N}$: size of population, $\boldsymbol{f}$: fitness function, $\boldsymbol{R_C}$: crossover probability, $\boldsymbol{R_M}$: mutation probability

Generate a random population $\boldsymbol{P}$ of $\boldsymbol{N}$ solutions $\boldsymbol{x^{(1)}}, \boldsymbol{x^{(2)}}, \dots, \boldsymbol{x^{(N)}}$ where $\boldsymbol{x^{(i)}} = \left(\boldsymbol{x_1^{(i)}},\ \boldsymbol{x_2^{(i)}}, \dots, \boldsymbol{x_D^{(i)}}\right)$ and $\boldsymbol{x_1^{(i)}} \in \{\boldsymbol{0}, \boldsymbol{1}\}$

$\boldsymbol{t} \leftarrow \boldsymbol{0}$ # Number of iterations (generations)

Repeat

Initialize a new population $\boldsymbol{P_{new}} \leftarrow \emptyset$;

For $\boldsymbol{i} \leftarrow \boldsymbol{1}$ **to** $\boldsymbol{N}$ **DIV 2 do**

Selection

Randomly select two solutions $\boldsymbol{x^{(P1)}}$ and $\boldsymbol{x^{(P2)}}$ from $\boldsymbol{P}$ as the first and second parents;

Crossover

Generate a random number $\boldsymbol{r} \in [\boldsymbol{0}, \boldsymbol{1}]$;

IF $(\boldsymbol{r} < \boldsymbol{R_c})$ **THEN**

Choose a random crossover point $\boldsymbol{k}$;

$\boldsymbol{x^{(C1)}} \leftarrow \boldsymbol{x^{(P1)}}[\boldsymbol{0}: \boldsymbol{k}] + \boldsymbol{x^{(P2)}}[\boldsymbol{k}:]$; $\quad \boldsymbol{x^{(C2)}} \leftarrow \boldsymbol{x^{(P2)}}[\boldsymbol{0}: \boldsymbol{k}] + \boldsymbol{x^{(P1)}}[\boldsymbol{k}:]$;

ELSE

$\boldsymbol{x^{(C1)}} \leftarrow \boldsymbol{x^{(P1)}}$; $\quad \boldsymbol{x^{(C2)}} \leftarrow \boldsymbol{x^{(P2)}}$;

End For;

Mutation

For $\boldsymbol{j} \leftarrow \boldsymbol{1}$ **to** $\boldsymbol{D}$ **do**

Generate a random number $\boldsymbol{r} \in [\boldsymbol{0}, \boldsymbol{1}]$;

IF $(\boldsymbol{r} < \boldsymbol{R_M})$ **THEN** $\boldsymbol{x^{(C1)}}[\boldsymbol{j}] \leftarrow \boldsymbol{1} - \boldsymbol{x^{(C1)}}[\boldsymbol{j}]$; **End IF;**

Generate a random number $\boldsymbol{r} \in [\boldsymbol{0}, \boldsymbol{1}]$

IF $(\boldsymbol{r} < \boldsymbol{R_M})$ **THEN** $\boldsymbol{x^{(C2)}}[\boldsymbol{j}] \leftarrow \boldsymbol{1} - \boldsymbol{x^{(C2)}}[\boldsymbol{j}]$; **End IF;**

End For;

Add $\boldsymbol{x^{(C1)}}$ and $\boldsymbol{x^{(C2)}}$ to $\boldsymbol{P_{new}}$;

End For;

#Replacement

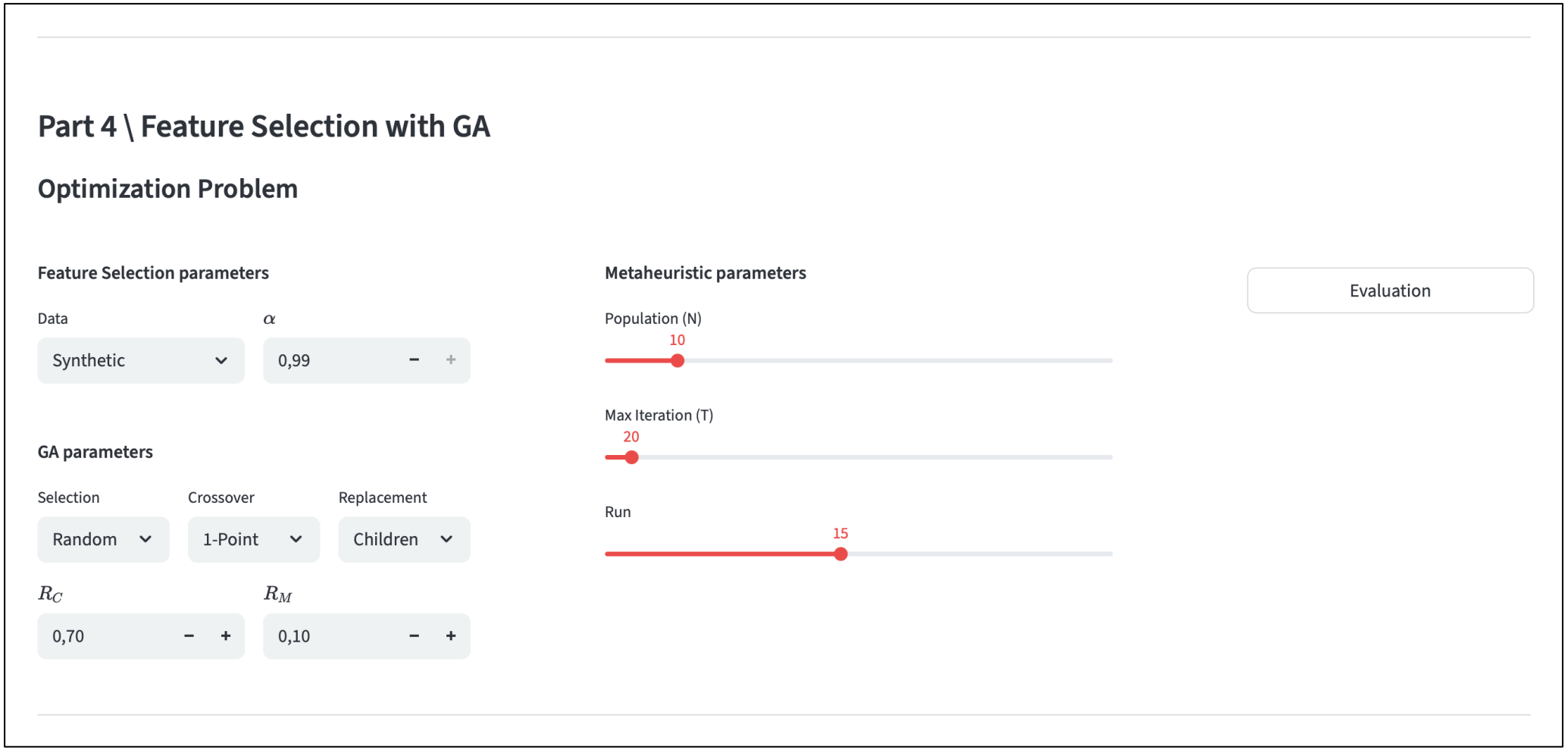
$\boldsymbol{P} \leftarrow \boldsymbol{P_{new}}$;

Evaluate the population $\boldsymbol{P}$ and update the best solution;

$\boldsymbol{t} \leftarrow \boldsymbol{t} + \boldsymbol{1}$

Until $(\boldsymbol{t} = \boldsymbol{T})$ **or** (~~$\boldsymbol{x^*}$ did not change during the last three iterations~~)

II. GA for feature selection



Feature Selection parameters

Data: Synthetic

α: 0,99

GA parameters

Selection: Random

Crossover: 1-Point

Replacement: Children

R_C: 0,70

R_M: 0,10

Metaheuristic parameters

Population (N): 10

Max Iteration (T): 20

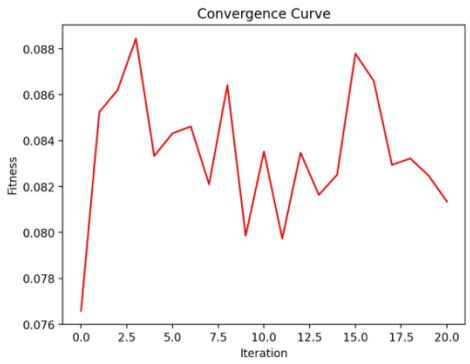
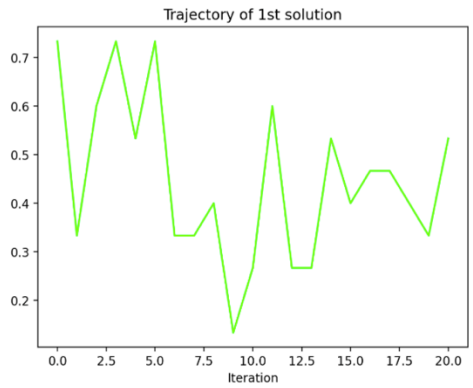
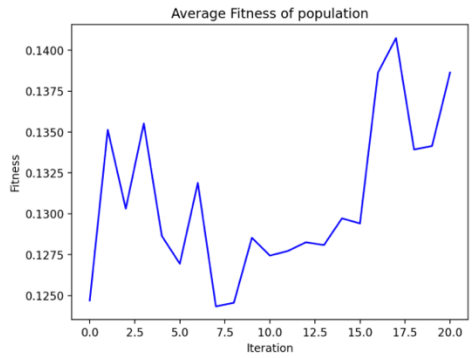
Run: 15

Evaluation

Best — 0.0636,

Mean (average error) — 0.0814,

Accuracy — 0.94,

Selected — 21,

STD — 0.0135,